

COUNTERMOVEMENT JUMP RELIABILITY PERFORMED WITH AND WITHOUT AN ARM SWING IN NCAA DIVISION 1 INTERCOLLEGIATE BASKETBALL PLAYERS

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ABSTRACT

Heishman, AD, Daub, BD, Miller, RM, Freitas, EDS, Frantz, BA, and Bemben, MG. Countermovement jump reliability performed with and without an arm swing in NCAA Division 1 intercollegiate basketball players. *J Strength Cond Res* XX(X): 000–000, 2018—The countermovement jump (CMJ) is routinely used in athlete performance to quantify adaptations to training, as well as monitor neuromuscular readiness and fatigue. However, controversy remains in whether to incorporate an arm swing during the CMJ (CMJ AS) or keep the hands placed on the hips (CMJ NAS). Incorporating the arms provides a higher degree of sport-specificity that may yield improved reliability, especially in skilled jumpers. By contrast, the hands-on-hips approach isolates lower extremity force production and eliminates potential arm-swing variation. Therefore, the purpose of this study was to establish the reliability of CMJ typical (CMJ-TYP), CMJ concentric alternative (CMJ-Conc-ALT), and CMJ eccentric alternative (CMJ-Ecc-ALT) variables obtained during the CMJ AS and CMJ NAS. Twenty-two (men = 14, women = 8) NCAA Division 1 collegiate basketball players performed 3 CMJ AS and 3 CMJ NAS on a force plate, in a randomized order. To assess the test-retest reliability, participants returned 1 week later to perform 3 additional CMJ AS and 3 CMJ NAS. Intraclass correlation coefficient (ICC) and coefficient of variation (CV) were used to assess intersession and intrasession reliability for the various CMJ variables. A majority of CMJ-TYP and several CMJ-Conc-ALT and CMJ-Ecc-ALT variables exhibited adequate intersession and intrasession reliability (ICC > 0.700 and CV < 10%) during both the CMJ AS and the CMJ NAS. Countermovement jump AS may provide more pertinent information about long-term changes in sport-specific performance, whereas the CMJ

NAS may be more beneficial for detecting acute changes in neuromuscular fatigue and athlete readiness.

KEY WORDS athlete monitoring, performance testing, smallest worthwhile change, neuromuscular readiness, vertical jump

INTRODUCTION

Effective athlete monitoring strategies should provide reliable information in a time-efficient manner while being minimally invasive to reduce undue stress or fatigue on the athlete (5,15,18). The countermovement jump (CMJ) is routinely used in high-performance sport settings to monitor changes in neuromuscular readiness, fatigue, and subsequent recovery in response to training loads, as well as to quantify adaptations to training programs (6,13–15,18,23). This information has also been coupled with internal and external load monitoring to provide a comprehensive insight into an athlete's readiness and response to training stimuli (1,18). Although the utilization of the CMJ to evaluate acute and long-term changes in performance indices is well-documented, previous literature lacks a universal consensus on the most effective testing methodology that produces the most reliable results.

Two methods are commonly used when performing the CMJ. The first approach incorporates the use of the arm swing (CMJ AS) (28,29,33,41,43). Several authors suggest that including an AS while performing a CMJ test yields a higher degree of sport specificity that may improve reliability, especially in skilled jumpers, because of an increased familiarity of the movement (33,36,39). Skilled jumpers have demonstrated the capacity to more effectively incorporate the AS during the CMJ (25,28). Furthermore, previous literature has demonstrated a relative consistency in the utilization of the AS during the CMJ in skilled jumpers, therefore introducing minimal variation to testing results (17,28,36,39).

The second approach when testing the CMJ eliminates the influence of the AS on CMJ performance (CMJ NAS), by requiring the athlete to maintain hand placement on their

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TABLE 1. Description of the countermovement jump typical variables.

CMJ typical variables	Abbreviation	Description
Concentric mean force [N]	ConcMF	Mean force during the concentric phase
Concentric mean power [W]	ConcMP	Mean power during the concentric phase
Concentric peak force [N]	ConcPF	Greatest force achieved during the concentric phase
Flight time [ms]	FT	Time spent in the air from jump take-off to landing
Flight time:contraction time	FT:CT	Ratio of flight time-to-contraction time
Jump height (flight time) [cm]	JH_FT	Maximal jump height computed using flight time
Jump height (imp-mom) [cm]	JH_Imp	Maximal jump height computed using impulse-momentum method
Peak power [W]	PP	Greatest power achieved

CMJ = countermovement jump; imp-mom = impulse-momentum method

hips or fixed to a virtually weightless implement (i.e., polyvinyl chloride pipe or wood dowel) positioned on their shoulders throughout the duration of the CMJ (1,4,6,12–14,23). Classic and contemporary literature rationalizes the elimination of the AS by suggesting it isolates lower extremity force production while mitigating the potential variations encountered with the AS movement (6,7,10,13). Indeed, previous literature has identified the CMJ to provide good reliability in typically reported metrics of jump height, force, and power (21). However, the unfamiliar nature of eliminating the AS during the CMJ has been speculated to alter movement strategies and thus change repeatability of the test, especially in athletes that participate in sports requiring a large quantity of jumping (25). Although CMJ AS and CMJ NAS share a variety of mechanical similarities, these complex movements exhibit markedly different mechanical events (34). Previous literature has

demonstrated altered movement strategies when performing either the CMJ approach, of which are likely to impact the force-time signature of each jumping strategy (11–14,28). Therefore, the alteration in movement strategy when eliminating the AS, combined with previous literature identifying sport-specific variations in jump strategy (28), bolsters the notion that incorporating the AS may provide insights into neuromuscular function more closely paralleling sport-specific tasks. However, the limited data specifically assessing the reliability of the CMJ with respect to the inclusion of the AS warrant further investigation.

The CMJ uses the propagation of a dynamic muscle action referred to as the stretch-shortening cycle (SSC) (26). Despite SSC activity requiring both concentric and eccentric muscle action, traditional CMJ variables of interest have centered around the components of the concentric phase or total jump outputs (i.e., jump height, force, and power),

TABLE 2. Description of the countermovement jump concentric alternative variables.

CMJ alternative variable	Abbreviation	Description
Concentric duration [ms]	ConcDur	Duration of the concentric phase
Concentric impulse [Ns]	ConcImp	Concentric force exerted multiplied by time taken
Concentric maximum RFD [$N \cdot s^{-1}$]	MaxRFD	Greatest rate of force increase during a 50-ms epoch over the concentric phase
Concentric peak velocity [$m \cdot s^{-1}$]	ConcPV	Greatest velocity achieved during the concentric phase
Concentric RFD [$N \cdot s^{-1}$]	ConcRFD	Rate of force development achieved over the concentric phase
Concentric RPD [$W \cdot s^{-1}$]	ConcRPD	Rate of power development from the start of the concentric phase to peak power
Contraction time [ms]	CT	Duration from jump initiation to take-off
Force at peak power [N]	F@PP	Force exerted at peak power
Reactive Strength Index modified [$m \cdot s^{-1}$]	RSI mod	Jump height (calculated from flight time) divided by contraction time
Velocity at peak power [$m \cdot s^{-1}$]	V@PP	Vertical velocity achieved at peak power during the take-off phase

CMJ = countermovement jump; ms = milliseconds; Ns = newtons; RFD = rate of force development

TABLE 3. Description of the countermovement jump eccentric alternative variables.

CMJ alternative variables	Abbreviation	Description
Braking phase duration [s]	BrakPDur	Period from minimum force to the start of the concentric phase
Eccentric braking RFD [$\text{N} \cdot \text{s}^{-1}$]	EccBRFD	Rate of force development from the minimum force at the start of the active braking phase to zero velocity at the end of the eccentric phase
Eccentric deceleration phase duration [s]	EccDPD	Time period from maximum negative velocity to zero velocity at the end of the eccentric phase
Eccentric deceleration RFD [$\text{N} \cdot \text{s}^{-1}$]	EccDecRFD	Rate of force development from the greatest negative velocity to zero velocity at the end of the eccentric phase
Eccentric duration [ms]	EccDur	Duration of the eccentric phase
Eccentric mean braking force [N]	EccMBF	Mean force during the eccentric braking phase from minimum force to zero velocity at the end of the eccentric phase
Eccentric mean deceleration force [N]	EccMDecF	Mean force during the eccentric braking phase from minimum force to zero velocity
Eccentric mean force [N]	EccMF	Mean force during the eccentric phase from start of movement to zero velocity
Eccentric mean power [W]	EccMP	Mean power during the eccentric phase from start of movement to zero velocity
Eccentric peak force [N]	EccPF	Greatest force achieved during the eccentric phase
Force at zero velocity [N]	F@0V	Force exerted at concentric onset
Lower-limb stiffness [$\text{N} \cdot \text{m}^{-1}$]	LLS	Change in vertical force divided by displacement of the countermovement during the eccentric phase

CMJ = countermovement jump; RFD = rate of force development

overlooking the performance of the eccentric components (6,7,10,12–14). Moreover, the complex and multifaceted changes associated with neuromuscular fatigue have catalyzed the exploration of CMJ variables beyond that of the traditional CMJ tests of jump height, force, and power. Examining these alternative CMJ variables may illuminate specific changes in neuromuscular function, such as those associated with the eccentric phase, which are not directly obtained in typical CMJ analyses focused on gross jump outcomes (13). Therefore, establishing the reliability of each CMJ phase may be critical for effectively monitoring changes in neuromuscular status as well as indices for clinicians and performance practitioners.

Although previous literature has explored the reliability of the CMJ, a paucity of literature exists directly comparing the reliability of the CMJ AS and CMJ NAS. More specifically, previous literature has yet to characterize the reliability of the alternative CMJ variables during a CMJ AS and CMJ NAS, especially in skilled jumpers. Therefore, the purpose of this study was to establish the intrasession and intersession reliability of the typical and alternative CMJ performance variables obtained from a force plate during a single CMJ when performed with an AS (CMJ) and when performed without an AS (CMJ NAS), in a cohort of NCAA Division 1 intercollegiate basketball players. It was hypothesized that most variables observed would meet the acceptable absolute

and relative intersession and intrasession reliability criterion during both the CMJ AS and CMJ NAS.

METHODS

Experimental Approach to the Problem

A randomized cross-over within subject study design was used to examine the reliability of both the CMJ AS and the CMJ NAS. Subjects performed 3 CMJs with an AS and 3 CMJs with no AS, in a randomized order, separated by at least 1 week. The dependent variables of interest, which includes typical CMJ variables, alternative CMJ variables related to the concentric movement, and alternative CMJ variables related to the eccentric movement, are outlined in Tables 1, 2 and 3, respectively. These variables will be explained in the Methods section.

Subjects

A convenience sample of 22 NCAA Division 1 collegiate basketball players (mean $\pm$ SD: men: $n = 14$, age = 19.7 ± 1.0 years [age range: 18–22 years], height = 197.6 ± 7.1 cm, body mass = 94.7 ± 6.2 kg; women: $n = 8$, age = 20 ± 1.6 years, height = 180.2 ± 6.5 cm, body mass = 78.2 ± 8.3 kg) was included in this study. All subjects were active squad members of the University of Oklahoma's men's and women's basketball teams. This research was approved by the Institutional Review Board of the University of Oklahoma, and all subjects provided written, informed consent before participating in the study.

TABLE 4. Countermovement jump typical variables intersession reliability.*

CMJ-typical variables	Arm swing (CMJ AS)											
	Intersession reliability						No arm swing (CMJ NAS)					
	Mean	Cron	ICC	TE	SWC	CV%	Mean	Cron	ICC	TE	SWC	CV%
Concentric mean force [N]	1,832.9	0.984	0.913	22.4	64.3	4.5	1,756.5	0.991	0.949	5.5	61.0	3.6
Concentric mean power [W]	2,809.9	0.974	0.862	58.3	157.5	9.2	2,554.6	0.992	0.957	14.6	131.4	5.5
Concentric peak force [N]	2,392.1	0.992	0.956	13.7	90.5	4.0	2,163.9	0.988	0.932	17.0	80.5	4.4
Flight time [ms]	595.3	0.987	0.927	2.3	12.8	2.6	541.7	0.993	0.957	3.1	11.9	2.4
Flight time:contraction time	0.76	0.887	0.566	0.03	0.04	16.2	0.66	0.955	0.778	0.02	0.03	9.3
Jump height (flight time) [cm]	43.9	0.987	0.927	0.2	1.8	5.1	36.40	0.993	0.958	0.4	1.6	4.7
Jump height (imp-mom) [cm]	44.5	0.883	0.556	1.7	2.6	12.7	35.77	0.989	0.939	0.4	1.61	5.4
Peak power [W]	5,827.8	0.968	0.834	99.2	296.2	7.9	4,735.8	0.993	0.961	17.0	206.2	3.7

*CMJ = countermovement jump; AS = arm swing; NAS = no arm swing; Cron = Cronbach's alpha; ICC = intraclass correlation coefficient; TE = typical error; SWC = smallest worthwhile change; CV% = coefficient of variation.

Procedures

All testing took place within a 2-week time frame during the off-season training period. Subjects performed 2 trial sessions of jump assessments performing 3 CMJ AS and 3 CMJ NAS.

The order of jump type was randomly assigned, and subjects performed the jump type in the reciprocal order during trial 2. A minimum of 2-minute rest was allotted between jump trials. In addition, each subject performed both testing

TABLE 5. Countermovement jump typical variables intrasession reliability.*

CMJ-typical variables	Intrasession reliability											
	Arm swing (CMJ AS)						No arm swing (CMJ NAS)					
	Mean	Cron	ICC	TE	SWC	CV%	Mean	Cron	ICC	TE	SWC	CV%
Concentric mean force [N]	1,832.9	0.973	0.923	89.6	64.5	3.8	1,756.5	0.988	0.965	53.6	61.2	2.8
Concentric mean power [W]	2,809.9	0.960	0.888	285.9	158.07	7.5	2,554.6	0.989	0.968	106.2	131.8	4.3
Concentric peak force [N]	2,392.1	0.989	0.966	84.0	90.8	3.1	2,163.9	0.978	0.937	84.5	80.7	3.9
Flight time [ms]	595.3	0.990	0.972	10.4	12.8	1.6	541.71	0.989	0.966	9.5	12.0	1.9
Flight time:contraction time	0.76	0.822	0.762	0.14	0.04	12.5	0.66	0.920	0.873	0.06	0.03	8.3
Jump height (flight time) [cm]	43.9	0.991	0.974	1.4	1.8	3.3	36.4	0.989	0.967	1.3	1.6	3.8
Jump height (imp-mom) [cm]	44.5	0.821	0.608	7.6	2.5	8.6	35.8	0.985	0.955	1.6	1.6	4.3
Peak power [W]	5,827.8	0.953	0.869	548.9	296.9	5.7	4,735.8	0.992	0.975	150.4	206.8	2.9

*CMJ = countermovement jump; AS = arm swing; NAS = no arm swing; Cron = Cronbach's alpha; ICC = intraclass correlation coefficient; TE = typical error; SWC = smallest worthwhile change; CV% = coefficient of variation.

TABLE 6. Countermovement jump alternative concentric variables intersession reliability.*

CMJ alternative variables	Intersession reliability											
	Arm swing (CMJ AS)						No arm swing (CMJ NAS)					
	Mean	Cronbach	ICC	TE	SWC	CV%	Mean	Cronbach	ICC	TE	SWC	CV%
Concentric duration [ms]	289.1	0.887	0.567	11.7	14.6	11.8	278.9	0.928	0.684	6.6	9.7	7.9
Concentric impulse [Ns]	265.6	0.965	0.820	4.6	11.5	6.2	238.5	0.996	0.976	1.4	8.9	2.7
Concentric maximum RFD [$\text{N} \cdot \text{s}^{-1}$]	8,580.9	0.958	0.791	455.1	914.8	23.3	4,716.9	0.477	0.132	344.7	471.8	31.9
Concentric peak velocity [$\text{m} \cdot \text{s}^{-1}$]	3.0	0.911	0.630	0.05	0.08	5.8	2.8	0.990	0.944	0.02	0.06	2.4
Concentric RFD [$\text{N} \cdot \text{s}^{-1}$]	3,637.6	0.954	0.776	253.7	495.26	32.8	2,161.2	0.854	0.494	191.1	357.1	63.9
Concentric RPD [$\text{W} \cdot \text{s}^{-1}$]	27,239.7	0.966	0.827	963.2	2,028.14	13.9	23,875.8	0.971	0.847	223.4	1,779.4	12.6
Contraction time [ms]	833.5	0.857	0.860	22.1	40.43	15.7	840.7	0.904	0.611	11.8	25.1	8.5
Force at peak power [N]	2,185.1	0.993	0.957	8.1	78.78	3.7	1,915.7	0.986	0.923	5.2	60.7	4.2
RSI mod [$\text{m} \cdot \text{s}^{-1}$]	0.57	0.935	0.707	0.03	0.04	17.3	0.46	0.972	0.854	0.01	0.03	10.4
Velocity at peak power [$\text{m} \cdot \text{s}^{-1}$]	2.65	0.901	0.604	0.05	0.07	6.3	2.5	0.986	0.921	0.02	0.05	3.0

*CMJ = countermovement jump; AS = arm swing; NAS = no arm swing; Cron = Cronbach's alpha; ICC = intraclass correlation coefficient; TE = typical error; SWC = smallest worthwhile change; CV% = coefficient of variation.

sessions within the same hour, and all subjects performed their testing in the afternoon between 13:00 and 14:00, as previous literature has identified the influence of time of day on jump performance (19). In accordance with previous literature (6) and in an attempt to control the impact of training loads on testing outcomes, CMJ testing was performed within the same time frame of the training week, and training loads were strictly matched 72 hours before both testing sessions, with sport-specific practice duration matched in the days before both trial sessions. Furthermore, subjects were instructed to have no physical exertion before arriving on days of testing. In an effort to maintain ecological validity, subjects wore their standard practice gear, including shoes of their choosing, and each subject was required to wear the same pair of shoes during both testing sessions. Furthermore,

no dietary restrictions were implemented; however, athletes were instructed to maintain normal dietary intake, as outlined by the team's sports nutritionist.

All testing was conducted at the basketball training facility before the start of strength training sessions. The same standardized warm-up was performed before each testing session, which included dynamic stretching and locomotion patterns (i.e., skipping, jogging, and running), similar to that of previous literature (6,18,19). Movement intensities gradually increased over the warm-up duration to prepare participants for maximal performance during the jump testing. Countermovement jumps were performed on the ForceDecks FD4000 Dual Force Platforms hardware (ForceDecks, London, United Kingdom), with a sample rate of 1,000 Hz. ForceDecks software (ForceDecks, London, United

TABLE 7. Countermovement jump alternative concentric variables intrasession reliability.*

	Intrasession average											
	Arm swing (CMJ AS)						No arm swing (CMJ NAS)					
	Mean	Cron	ICC	TE	SWC	CV%	Mean	Cron	ICC	TE	SWC	CV%
Concentric duration [ms]	289.1	0.830	0.656	42.2	14.5	9.7	278.9	0.901	0.753	21.9	9.7	6.3
Concentric impulse [Ns]	265.6	0.945	0.850	22.4	11.5	4.4	238.5	0.996	0.987	5.4	8.9	2.2
Concentric maximum RFD [$\text{N} \cdot \text{s}^{-1}$]	8,482.2	0.920	0.792	1,922.8	918.1	20.0	4,662.5	0.231	0.103	1,832.2	467.0	32.5
Concentric peak velocity [$\text{m} \cdot \text{s}^{-1}$]	3.0	0.859	0.671	0.2	0.1	4.1	2.8	0.986	0.958	0.1	0.1	1.9
Concentric RFD [$\text{N} \cdot \text{s}^{-1}$]	3,637.6	0.919	0.798	1,136.8	497.2	30.7	2,144.3	0.662	0.395	1,177.1	354.6	54.5
Concentric RPD [$\text{W} \cdot \text{s}^{-1}$]	28,635.3	0.955	0.879	3,016.0	2,341.5	9.9	23,875.8	0.959	0.885	2,576.9	1,781.5	9.9
Contraction time [ms]	833.5	0.783	0.545	144.4	40.5	12.7	840.7	0.843	0.644	71.3	25.2	7.6
Force at peak power [N]	2,185.1	0.990	0.970	66.9	79.1	3.1	1,915.7	0.981	0.945	58.7	60.9	3.6
RSI mod [$\text{m} \cdot \text{s}^{-1}$]	0.57	0.893	0.738	0.1	0.04	12.8	0.46	0.956	0.879	0.04	0.03	9.1
Velocity at peak power [$\text{m} \cdot \text{s}^{-1}$]	2.6	0.838	0.635	0.2	0.1	4.6	2.5	0.980	0.943	0.1	0.1	2.5

*CMJ = countermovement jump; AS = arm swing; NAS = no arm swing; Cron = Cronbach's alpha; ICC = intraclass correlation coefficient; TE = typical error; SWC = smallest worthwhile change; CV% = coefficient of variation.

Kingdom) was used to analyze all CMJs and generate the CMJ variables. The ForceDecks software uses a 20N offset from the measured body mass, quantified before the jump, to define the start of movement. The end of eccentric and start of concentric was defined as minimum displacement (absolute) that is equal to zero velocity, whereas take-off was defined as the timepoint at which total vertical force fell below the threshold of 20N below body mass. A description of the CMJ typical variables, CMJ concentric alternative variables, and CMJ eccentric alternative are outlined in Table 1, Table 2, and Table 3, respectively. The variables included for analysis were selected because they are part of the ForceDecks Software default analysis output and may be of interest to practitioners. A multitude of variables were incorporated in the analysis because, as previous literature has suggested, the most reliable variables may not be the most efficacious in monitoring and detecting neuromuscular fatigue or changes in athlete performance (6). Furthermore, variables were characterized as typical and alternative based on similar methods of previous literature (6,13). To limit the impact of instructions on the CMJ performance character-

istics, consistent instructions were provided to all participants during each CMJ trial (42). In addition, verbal encouragement was provided to support maximal effort during each jump attempt.

Countermovement Jump With an Arm Swing. Subjects started in the tall standing position, with feet placed hip width to shoulder width apart, but with hands free for movement. The subject was then instructed to start with equal weight distribution on both force cells. A visual representation of weight distribution was displayed on a monitor in front of the participant to provide synchronized and integrated feedback, allowing for the subject to adjust their positioning for equal quantities of body mass to be distributed on each force cell for the start of the jump. The subject then dropped into the countermovement position to a self-selected depth, incorporating an AS in their most natural, self-selected manner, followed by a maximal effort vertical jump and landing in an athletic position on the force cells. The participant resets to the starting position after each jump, and the procedure was completed for a total of 3 jumps. If at

TABLE 8. Countermovement jump alternative eccentric variables intersession reliability.*

CMJ alternative variables	Intersession reliability											
	Arm swing (CMJ AS)						No arm swing (CMJ NAS)					
	Mean	Cronbach	ICC	TE	SWC	CV%	Mean	Cronbach	ICC	TE	SWC	CV%
Braking phase duration [s]	0.33	0.799	0.398	0.02	0.03	25.5	0.34	0.343	0.488	0.01	0.02	15.3
Eccentric braking RFD [N·s ⁻¹]	5,293.3	0.967	0.830	312.4	599.6	23.3	5,107.8	0.968	0.835	208.6	590.3	26.1
Eccentric deceleration phase duration [s]	0.19	0.831	0.450	0.01	0.02	21.0	0.18	0.872	0.531	0.01	0.01	17.3
Eccentric deceleration RFD [N·s ⁻¹]	6,365.1	0.973	0.859	283.8	840.1	27.0	6,551.4	0.973	0.858	306.3	857.4	26.3
Eccentric duration [ms]	544.5	0.754	0.338	26.9	38.7	26.2	561.8	0.868	0.522	7.5	21.0	11.7
Eccentric mean braking force [N]	1,070.5	0.960	0.799	19.2	37.2	6.7	1,054.5	0.977	0.875	10.5	37.2	5.7
Eccentric mean deceleration force [N]	1,466.7	0.978	0.881	37.2	69.6	8.1	1,473.1	0.975	0.865	34.8	70.4	9.0
Eccentric mean force [N]	884.1	0.994	0.964	3.0	22.2	1.8	882.6	0.999	0.995	0.4	22.7	0.8
Eccentric mean power [W]	489.7	0.881	0.552	36.2	41.7	24.7	507.5	0.936	0.710	17.8	34.5	17.9
Eccentric peak force [N]	1,836.0	0.970	0.842	55.1	89.2	8.9	1,953.7	0.973	0.858	32.5	95.6	9.5
Force at zero velocity [N]	1,575.5	0.995	0.973	13.8	160.40	6.9	1,616.2	0.994	0.966	29.5	170.7	7.6
Lower-limb stiffness [N·m ⁻¹]	9,940.9	0.220	0.045	3,795.1	5,805.0	59.4	5,473.6	0.745	0.328	193.2	663.2	27.0

*CMJ = countermovement jump; AS = arm swing; NAS = no arm swing; Cron = Cronbach's alpha; ICC = intraclass correlation coefficient; TE = typical error; SWC = smallest worthwhile change; CV% = coefficient of variation; RFD = rate of force development.

any point the subject exhibited excessive knee flexion once airborne, the jump was ruled invalid and repeated.

Countermovement Jump with No Arm Swing. In the same manner as the CMJ AS, subject started in the tall standing position, with feet placed hip width to shoulder width apart and hands akimbo. The subject was then instructed to start with equal weight distribution on both force cells. A visual representation of weight distribution was displayed on

a monitor in front of the participant to provide synchronized and integrated feedback, allowing for the participant to adjust their positioning for equal quantities of body mass to be distributed on each force cell for the start of the jump. Subjects then dropped into the countermovement position to a self-selected depth, followed by a maximal effort vertical jump, and landed in an athletic position on the force cells. The subject resets to the starting position after each jump, and the procedure was completed for a total of 3 jumps. If at

TABLE 9. Countermovement jump alternative eccentric variables intrasession reliability.*

	Intrasession average											
	Arm swing (CMJ AS)						No arm swing (CMJ NAS)					
	Mean	Cron	ICC	TE	SWC	CV%	Mean	Cron	ICC	TE	SWC	CV%
Braking phase duration [s]	0.33	0.741	0.492	0.11	0.03	20.1	0.34	0.840	0.637	0.05	0.02	12.6
Eccentric braking RFD [$\text{N} \cdot \text{s}^{-1}$]	5,253.5	0.927	0.808	1,273.3	581.2	21.6	5,107.8	0.932	0.824	1,126.3	592.7	22.7
Eccentric deceleration phase duration [s]	0.20	0.831	0.678	0.05	0.02	15.9	0.18	0.789	0.563	0.03	0.01	14.2
Eccentric deceleration RFD [$\text{N} \cdot \text{s}^{-1}$]	6,041.9	0.952	0.869	1,353.5	784.2	20.7	6,551.4	0.861	0.707	1,480.4	860.7	23.5
Eccentric duration [ms]	544.5	0.658	0.391	156.4	38.5	22.0	561.8	0.856	0.703	68.4	21.0	11.1
Eccentric mean braking force [N]	1,070.4	0.921	0.798	93.0	37.1	5.5	1,054.5	0.941	0.843	67.1	37.3	5.0
Eccentric mean deceleration force [N]	1,466.7	0.964	0.901	109.4	69.9	6.8	1,473.1	0.948	0.857	122.0	70.7	8.0
Eccentric mean force [N]	884.0	0.988	0.966	19.7	22.3	1.4	882.6	0.999	0.999	4.6	22.8	0.4
Eccentric mean power [W]	489.7	0.833	0.628	124.8	41.6	22.5	507.5	0.912	0.776	71.8	34.4	15.9
Eccentric peak force [N]	1,836.0	0.952	0.874	152.2	89.4	7.8	1,953.7	0.959	0.901	153.8	95.9	8.3
Force at zero velocity [N]	1,498.4	0.989	0.968	123.0	152.6	5.8	1,616.2	0.991	0.972	139.1	171.4	6.4
Lower-limb stiffness [$\text{N} \cdot \text{m}^{-1}$]	9,986.2	0.348	0.153	18,644.4	4,536.5	32.7	5,473.6	0.562	0.319	2,108.4	642.1	21.4

*CMJ = countermovement jump; AS = arm swing; NAS = no arm swing; Cron = Cronbach's alpha; ICC = intraclass correlation coefficient; TE = typical error; SWC = smallest worthwhile change; CV% = coefficient of variation; RFD = rate of force development.

any point the subject removed their hands from their hips or exhibited excessive knee flexion once airborne, the jump was ruled invalid and repeated.

Statistical Analyses

Intersession and intrasession means and reliability were computed for the typical and alternative variables of the CMJ when performed with an AS, as well as when performed without an AS. Relative reliability was assessed using intraclass correlation coefficients (ICCs). Based on recommendation of previous literature, obtaining an ICC ≥ 0.70 was set as a minimum acceptable reliability (3). In addition, absolute reliability was assessed using coefficient of variation (CV%) and typical error (TE) of measurement. Acceptable absolute reliability was established at CV $\leq 10\%$

(4,6,13). Nevertheless, previous literature has suggested that test reliability standards should ultimately be judged by the individual researcher or practitioner based in accordance with their intended use, and that the most reliable variables may not necessarily be the most efficacious in athlete monitoring and performance testing regimens (6). The TE of measurement was calculated by dividing the *SD* by the square root of 2 to provide a reflection of the noise within the test caused by biological and technical aspects (20). In addition, Cronbach's alpha was computed to assess and provide a measure of internal consistency, allowing for the present analysis to be compared with results that only determined Cronbach's alpha (38). Finally, based on suggestions of previous work, the smallest worthwhile change (SWC) was calculated as 0.2 X between-subject *SD* and

represents the smallest change that is of benefit for athletic performance (6,13,20). If the $TE \leq SWC$, the metrics were deemed capable of detecting the SWC (6,13,20). SPSS (version 24, Armonk, NY) was used for all data analysis.

RESULTS

Intersession results for CMJ typical variables during both CMJ AS and CMJ NAS are outlined in Table 4 while intrasession results are in Table 5. Most CMJ typical variables met the acceptable levels of intersession and intrasession relative ($ICC > 0.70$) and absolute ($CV\% < 10$) reliability during both the CMJ AS and CMJ NAS.

Intersession and intrasession results for CMJ concentric alternative variables are presented in Tables 6 and 7, respectively. Intersession and intrasession results for CMJ eccentric alternative variables are presented in Tables 8 and 9, respectively. During the CMJ AS 4 of 10 CMJ concentric alternative variables met the reliability criterion intersession, whereas 6 of 10 variables met the relative and absolute reliability criteria during the CMJ NAS. In addition, most of these metrics were capable of detecting meaningful intersession change. However, many variables failed to meet the established reliability criterion intrasession. Five of 12 CMJ eccentric alternative variables met the relative and absolute reliability criterion intersession and intrasession for both the CMJ AS and CMJ NAS. All tables show mean, Cronbach's alpha, TE, and %CV.

DISCUSSION

The purpose of the present investigation was to establish the intersession and intrasession reliability of the typical and alternative CMJ variables obtained during a single CMJ when performed with (CMJ AS) and without an AS (CMJ NAS) in a cohort of skilled jumpers. The primary findings of this study were (a) a variety of typical and alternative variables derived from the CMJ-expressed acceptable levels of intersession and intrasession reliability ($ICC > 0.70$; $CV < 10\%$) during both the CMJ AS and CMJ NAS; (b) intersession reliability analysis revealed an improved capacity to detect change ($SWC > TE$), as compared to intrasession reliability analysis; and (c) the enhanced absolute reliability advocates using CMJ NAS for monitoring acute changes in fatigue and neuromuscular status, whereas CMJ AS may be more advantageous for evaluating changes between training phases. The current observations reinforce previous literature indicating that the CMJ typical variables met acceptable intersession and intrasession criterion for both relative and absolute reliability during both jump protocols. Previous literature has consistently supported the reliability of the CMJ typical variables during the CMJ NAS (1,4,6,10,13,21,30) and the CMJ AS (17,22,32,40). In addition, the present investigation uniquely identifies the capability of the CMJ AS to yield similar acceptable levels of reliability in skilled jumpers.

Gathercole et al. (13) designed one of the first investigations to evaluate CMJ reliability beyond the CMJ typical

variables. The design evaluated CMJ NAS reliability, reporting 11 intersession and 16 intrasession variables exhibiting high levels of repeatability (with $CV < 5\%$). Their results identified higher reliability than many of the variables in this study, which may stem from a novel sampling approach that averaged the most similar jumps within a session in terms of power output and jump duration (13). Although this approach facilitated improvements in reliability, it may not be practical in applied practice for coaches wanting immediate feedback on jump testing. Recent work by Edwards et al. (10) explored CMJ NAS reliability in male collegiate basketball players and reported several CMJ typical variables displayed acceptable absolute intersession and intrasession reliability ($CV < 10\%$). Despite several CMJ typical variables meeting acceptable reliability criterion, Edwards et al. (10) reported no CMJ typical variables capable of detecting intersession or intrasession SWC ($SWC < TE$). By contrast, this study found that a majority of the metrics seem capable of detecting intersession SWC, whereas fewer metrics possessed the ability to detect intrasession SWC. In addition, their results outlined greater intersession variability for flight time:contraction time (FT:CT), as well as jump height (JH) when compared with the CMJ NAS findings of this study. The conflicting results between this study and those of Edwards et al. (10) may be associated with changes in training load around the time of testing because their attempt to control training load was not discussed in their methodology and has been shown to influence jump performance. For example, previous literature has demonstrated that increases in training load may elicit decrements in jump performance in collegiate basketball players (18). Particularly, Edwards et al. (10) noted intrasession mean JH of 36 cm and intersession mean JH of 33 cm, which may have exacerbated the marginal reliability observed in their analysis. Because of previous literature postulating the residual effect of training that may influence CMJ performance lasting up to 72 hours (8), this study attempted to control for training load, which may support the improved intersession reliability observed.

One of the most documented variables of the CMJ is the estimation of JH. A variety of technology have been used to assess JH, such as a Vertec and contact mats making JH computation readily accessible (18,22,36,43). In this study, JH was calculated both through the flight time method and the impulse-momentum method (33). For both CMJ protocols, JH_FT showed high levels of both intersession and intrasession reliability. These findings support previous literature characterizing high levels of JH_FT reliability, supporting its use in applied practice (4,6,36,41). In contrast to JH_FT, JH_Imp demonstrated a high level of reliability during the CMJ NAS protocol but failed to meet the relative intersession and intrasession reliability criteria during the CMJ AS. Although not a primary focus of this study, alteration in movement strategy resulting in increases in velocity at take-off during the CMJ AS may exaggerate variability in the velocity at take-off, introducing greater variability in the

JH_Imp computation (29). In addition, previous literature has suggested misidentifying the instant of take-off by as little as 2–3 milliseconds can manifest into a 2% variation in velocity and displacement (16,33). Although previous literature has characterized the impulse-momentum theorem as the “gold-standard” of flight time computation, the findings of this study found higher reliability of the FT calculation to estimate jump height during the CMJ AS protocol (33). Therefore, if using the CMJ NAS protocol, both JH_FT and JH_Imp computational techniques will provide reliable information capable of detecting change. When performing CMJ AS protocol, JH should be computed from flight time method rather than impulse-momentum method for enhanced reliability.

FT showed high levels of intersession and intrasession reliability during both CMJ AS and CMJ NAS, whereas FT:CT only met acceptable reliability criterion during the CMJ NAS. The present findings support that of previous literature suggesting acceptable intersession and intrasession reliability of FT:CT during the CMJ NAS (6,13). In this study, stable FT reliability potentially explains the superior reliability of JH_FT for both protocols. Therefore, low levels of FT:CT reliability during the CMJ AS are speculated to have resulted from the low levels of reliability for CT (CV = 15.7%) experienced during the CMJ AS. The same verbal cues of “jump as high as possible” and “jump as you naturally would” were given during each protocol. Therefore, a potential explanation for the variations in CT that influenced the reliability of FT:CT may allude to the athletes’ capacity to manipulate the AS to affect overall jump outcomes (13,28). However, to our knowledge, this is the first study to document FT:CT reliability during a CMJ AS in skilled jumpers.

This study offers novel insight into the variables of ConcRFD and MaxRFD. Although these variables failed to meet the absolute reliability criterion, both met the minimum intersession and intrasession relative reliability criterion during the CMJ AS, but not during the CMJ NAS. In addition, CV% was substantially reduced during the CMJ AS, as compared to CMJ NAS. Furthermore, these variables seem capable of detecting meaningful changes in intersession SWC. Previous research has consistently identified rate of force development (RFD) variables as unreliable, with ConcRFD trends ranging from CV = 17.9%–40% and MaxRFD ranging from CV = 16–36% (4,13,21,32). Although their analysis only examined CMJ AS, McLellan et al. (32) has documented some of the highest reliability values for RFD (MaxRFD ICC = 0.89, CV = 16.3%; ConcRFD ICC = 0.89, CV = 17.9%). The results of this study reinforce their findings and suggest RFD reliability may be improved during CMJ AS.

The RFD has been postulated as a key component of athletic endeavors (23,26,32). More specifically, basketball play requires the ability to generate force in an abbreviated time frame during maximal efforts of accelerations and decelerations in all 3 orthogonal planes (1,18). Therefore, quan-

titating changes in ConcRFD may be advantageous in determining the effectiveness of a training program. The present data reveal improved reliability metrics for RFD variables during the CMJ AS in skilled jumpers. A potential explanation may be extrapolated from evidence in elite volleyball players, which has illuminated a small variation in time when the upward AS begins, showing the high and consistent level of utilization of the arm swim during the CMJ AS (34). The improved reliability and perceptible capacity to detect meaningful intersession change may allow ConcRFD variables during the CMJ AS to detect longer term changes in performance, such as changes over various training phase when greater shifts in performance are expected. By contrast, these variables may be less useful in the assessment of acute changes in performance, such as monitoring changes in neuromuscular readiness and fatigue. Their low levels of absolute reliability will likely lead to the minor variations in performance falling within the error of measurement.

Similar to RFD, rate of power development (RPD) has been speculated as an efficacious and novel performance diagnosis variable but is rarely examined within the literature (7,21). In this study, ConcRPD showed similar intrasession reliability criterion for both CMJ AS and CMJ NAS meeting the absolute and relative criteria but fell just outside of the acceptable threshold for intersession reliability for both protocols. Cormie et al. (7) reported much higher test-retest reliability (ICC = 0.99) during a CMJ NAS protocol, whereas Hori et al. (21) observed high levels of reliability for RPD (ICC = 0.91 and CV = 8.2%), both congruent with the findings of Gathercole et al. (13) (interday CV = 7.3% and intraday CV = 4.6%). Interestingly, the data analysis performed by Gathercole et al. (12) may be more advantageous for “rate” variables, such that smoothing the variability will allow practitioners to identify meaningful change in variables that exhibit less reliability but may still foster a high pertinence to sport performance (13). This study also only examined single-measures ICC; however, using the average measure ICC may hoist an improvement in relative reliability for less reliability variables (4). According to suggestions from previous literature, increasing sampling rate beyond the 1,000 Hz used in this study may aid in reliability improvements of these variables; however, these methods may be implausible for practical application (21).

RSI_{mod} evaluates an athlete’s ability to create maximal vertical impulse in a minimal amount of time during the CMJ (2,9,37), has been shown to be a valid measure of lower-extremity explosiveness (24), and represents the distinct muscle action associated with the slow SSC (2,9,31,37). Previous literature has investigated the relative reliability of RSI_{mod} during CMJ NAS reporting a range of ICC from 0.850 to 0.960 (2,9,37). The present data support previous findings with RSI_{mod} exhibiting acceptable intersession and intrasession relative reliability during the CMJ NAS. Bailey et al. (2) assessed the absolute reliability of RSI_{mod}, reporting

a CV of 12.7%, nearly identical to that observed in the present analysis. However, when compared with the CMJ NAS, this study demonstrated higher levels of intersession and intrasession absolute reliability for RSI_{mod} . The higher levels of absolute reliability may be attributed to the varying sport demands of the 2 cohorts (2). Nonetheless, results from this study support the assessment of RSI_{mod} through the CMJ NAS approach in skilled jumpers.

ConcPV met the acceptable relative reliability criterion during the CMJ NAS, and meaningful change only appeared capable of being distinguished intersession. The findings of this study support those of Sheppard et al. (35), who observed low relative reliability ($ICC = 0.25$) but adequate absolute reliability ($CV = 7.3\%$) during a CMJ NAS protocol in a cohort of volleyball players. By contrast, previous work has shown CMJ NAS protocol to produce acceptable relative and absolute reliability of PV (4,13,21,29). Likewise, ConcDur, CT, and V@PP, only met acceptable relative and absolute reliability criteria during the CMJ NAS. These findings are in agreement with previous literature using the CMJ NAS approach (10,13). Ultimately, if the variables of interest are ConcPV, ConcDur, CT, or V@PP, the CMJ NAS is likely the superior choice to capture more reliable data.

The variables of EccMBF, EccMF, EccMP, and EccPF met the acceptable intersession and intrasession reliability criteria for both jump protocols. Although seldomly explored, Meylan et al. (34) observed similar levels of reliability for EccPF ($CV = 3.5\%$). Although improved reliability was observed during the CMJ NAS, the novel exploration of BrakDur, EccBRFD, Ecc DecDur, Ecc DRFD, EccMP, and lower-limb stiffness failed to meet the minimum intersession and intrasession reliability criteria. The present analysis parallels previous literature, in that eccentric RFD variables exhibits adequate relative reliability ($Cron = 0.960$; $ICC = 0.880$) but limited absolute reliability (4). As these variables are speculated to be associated with neuromuscular fatigue (13,14), future research should explore using the Gathercole et al. (13) approach of analysis, which may potentially improve the reliability of these variables. In addition, the present data support most of the eccentric variables to be adept in detecting intersession meaningful change SWC.

Intersession and intrasession analysis identified F@0V as adequately reliable during both jump protocols. Interestingly, absolute reliability was slightly improved during the CMJ as compared to the CMJ NAS. Previous literature also has identified F@0V as a reliable CMJ alternative variable (10,13,34). Edwards et al. (10) reported similar results of reliability adequacy as this study; however, their analysis seemed incapable of detecting meaningful change, while the present analysis both intersession and intrasession $SWC > TE$.

The present data and those of Edwards et al. (10) indicate that EccDur failed to meet reliability criterion; however, Gathercole et al. (13) identified EccDur as a reliable variable (inter $CV = 8.0\%$; intra $CV = 6.2\%$). A potential explanation

for their results of acceptable reliability may be associated with their novel approach in analysis as previously mentioned. It is also possible that skilled jumpers can manipulate the eccentric components of the jump, yielding greater variability in loading patterns, to accomplish the desired gross jump outcome (i.e., JH) (10,13,27). Therefore, understanding the underlying kinetics of the eccentric component may provide more pertinent information regarding neuromuscular function. However, it should be noted that EccDur repeatability was improved during the CMJ NAS, which suggests the CMJ NAS protocol may be more advantageous when eccentric components of the CMJ are of primary focus.

This study has limitations that warrant discussion. First, this study lacked the capacity to discretely quantify training load during the testing period, such as internal or external athlete monitoring strategies (15). However, the present analysis relied on matching time on the court because previous literature has demonstrated the relationship between external load and activity duration (18). Second, this study examined a relatively homogenous cohort of skilled jumpers with small between-subject variation in many jump parameters, which may have contributed a reduction in relative reliability. Finally, it should also be emphasized that the most reliable variables may not be the most efficacious in monitoring and detecting neuromuscular fatigue or changes in athlete performance. As previous literature has articulated, it is ultimately up to the scientist and/or practitioner to establish acceptable reliability criteria and deem a variable reliable enough for its intended use (6). Therefore, future research should explore the sensitivity and specificity of the various CMJ variables in this study in detecting alterations in neuromuscular fatigue and function.

PRACTICAL APPLICATIONS

The results of this study provide several key practical applications for assessing and monitoring athlete performance. Both CMJ AS and CMJ NAS provide reliable information associated with CMJ typical and alternative variables in skilled jumpers, but each protocol may offer distinct advantages depending on which variables are of interest. Coaches and practitioners should use the CMJ AS in the assessment of longer term changes in performance, such as alterations in performance across training phases. The CMJ NAS may be more useful in isolating lower-extremity function and quantitating acute changes in neuromuscular readiness and fatigue, such as an athlete monitoring strategy.

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REFERENCES

- Aoki, MS, Ronda, LT, Marcelino, PR, Drago, G, Carling, C, Bradley, PS, and et al. Monitoring training loads in professional basketball players engaged in a periodized training program. *J Strength Cond Res* 31: 348–358, 2017.
- Bailey, C, Suchomel, TJ, Beckham, GK, Sole, CJ, and Grazer, JL. A comparison of baseball positional differences with reactive strength index-modified. *Proc Int Conf Biomech Sport* 18–21, 2014. Available from: <https://ojs.ub.uni-konstanz.de/cpa/article/view/5994>.
- Baumgartner, TA and Chung, H. Confidence limits for intraclass reliability coefficients. *Meas Phys Educ Exerc Sci* 5: 179–188, 2001.
- Byrne, PJ, Moody, JA, Cooper, S-M, Kinsella, S, and Byrne, P. The reliability of countermovement jump performance and the reactive strength index in identifying drop-jump drop height in hurling players. *J Exerc Sport Med* 1: 1–10, 2017.
- Conlon, J, Haff, GG, Nimphius, S, Tran, T, and Newton, RU. Vertical jump velocity as a determinant of speed and agility performance. *J Aust Strength Cond* 21: 88–90, 2013.
- Cormack, SJ, Newton, RU, McGulgan, MR, and Doyle, TLA. Reliability of measures obtained during single and repeated countermovement jumps. *Int J Sports Physiol Perform* 3: 131–144, 2008.
- Cormie, P, McBride, JM, and McCaulley, GO. Power-time, force-time, velocity-time curve analysis of the countermovement jump impact of training 23: 177–186, 2009.
- Dousset, E, Avela, J, Ishikawa, M, Kallio, J, Kuitunen, S, Kyröläinen, H, et al. Bimodal recovery pattern in human skeletal muscle induced by exhaustive stretch-shortening cycle exercise. *Med Sci Sports Exerc* 39: 453–460, 2007.
- Ebben, WP and Petushek, EJ. Using the reactive strength index modified to evaluate plyometric performance. *J Strength Cond Res* 24: 1983–1987, 2010.
- Edwards, T, Spiteri, T, Piggott, B, Bonhotal, J, Haff, GG, and Joyce, C. Reliability and sensitivity of neuromuscular and perceptual fatigue measures in collegiate Men's basketball. *J Strength Cond Res*, 2018. Epub ahead of print.
- Feltner, ME, Bishop, EJ, and Perez, CM. Segmental and kinetic contributions in vertical jumps performed with and without an arm swing. *Res Q Exerc Sport* 75: 216–230, 2004.
- Gathercole, R, Sporer, B, and Stellingwerff, T. Countermovement jump performance with increased training loads in elite female rugby athletes. *Int J Sports Med* 36: 722–728, 2015.
- Gathercole, R, Sporer, B, Stellingwerff, T, and Sleivert, G. Alternative countermovement-jump analysis to quantify acute neuromuscular fatigue. *Int J Sports Physiol Perform* 10: 84–92, 2015.
- Gathercole, RJ, Sporer, BC, Stellingwerff, T, and Sleivert, GG. Comparison of the capacity of different jump and sprint field tests to detect neuromuscular fatigue. *J Strength Cond Res* 29: 2522–2531, 2015.
- Halsen, SL. Monitoring training load to understand fatigue in athletes. *Sport Med* 44: 139–147, 2014.
- Hara, M, Shibayama, A, Takeshita, D, Hay, DC, and Fukushima, S. A comparison of the mechanical effect of arm swing and countermovement on the lower extremities in vertical jumping. *Hum Mov Sci* 27: 636–648, 2008.
- Harman, EA, Rosenstein, M, Frykman, P, and Rosenstein, R. The effect of arms and countermovement on vertical jumping. *Med Sci Sports Exerc* 22: 825–833, 1990, 1990.
- Heishman, AD, Curtis, MA, Saliba, E, Hornett, RJ, Malin, SK, and Weltman, AL. Noninvasive assessment of internal and external player load. *J Strength Cond Res* 32: 1280–1287, 2018.
- Heishman, AD, Curtis, MA, Saliba, EN, Hornett, RJ, Malin, SK, and Weltman, AL. Comparing performance during morning vs. afternoon training sessions in intercollegiate basketball players. *J Strength Cond Res* 31: 1557–1562, 2017.
- Hopkins, WG, Marshall, SW, Batterham, AM, and Hanin, J. Progressive statistics for studies in sports medicine and exercise science. *Med Sci Sports Exerc* 41: 3–12, 2009.
- Hori, N, Newton, RU, Kawamori, N, McGuigan, MR, Kraemer, WJ, and Nosaka, K. Reliability of performance measurements derived from ground reaction force data during countermovement jump and the influence of sampling frequency. *J Strength Cond Res* 23: 874–882, 2009.
- Isaacs, LD. Comparison of the vertec and Just Jump Systems for measuring height of vertical jump by young children. *Percept Mot Skills* 86: 659–663, 1998.
- Kavanaugh, AA, Mizuguchi, S, Sands, WA, Ramsey, MW, and Stone, MH. Long-term changes in jump performance and maximum strength in a cohort of NCAA Division I women's volleyball athletes. *J Strength Cond Res* 32: 66–75, 2017.
- Kipp, K, Kiely, MT, and Geiser, CF. Reactive strength index modified is a valid measure of explosiveness in collegiate female volleyball players. *J Strength Cond Res* 30: 1341–1347, 2016.
- Klavora, P. Vertical-jump tests: A critical review. *Strength Cond J* 22: 70–75, 2000.
- Komi, PV. Stretch-shortening cycle: A powerful model to study normal and fatigued muscle. *J Biomech* 33: 1197–1206, 2000.
- Laffaye, G, Bardy, B, and Taiar, R. Upper-limb motion and drop jump: Effect of expertise. *J Sports Med Phys Fitness* 46: 238–247, 2006.
- Laffaye, G, Wagner, PP, and Tombleson, TIL. Countermovement jump height: Gender and sport-specific differences in the force-time variables. *J Strength Cond Res* 28: 1096–1105, 2014.
- Lees, A, Vanrenterghem, J, and Clercq, DD. Understanding how an arm swing enhances performance in the vertical jump. *J Biomech* 37: 1929–1940, 2004.
- Markovic, G, Dizdžar, D, Jukic, I, and Cardinale, M. Reliability and factorial validity of squat and countermovement jump tests. *J Strength Cond Res* 18: 551–555, 2004.
- Markwick, WJ, Bird, SP, Tufano, JJ, Seitz, LB, and Haff, GG. The intraday reliability of the reactive strength index calculated from a drop jump in professional men's basketball. *Int J Sports Physiol Perform* 10: 482–488, 2015.
- McLellan, CP, Lovell, DI, and Gass, GC. The role of rate of force development on vertical jump performance. *J Strength Cond Res* 25: 379–385, 2011.
- McMahon, JJ, Suchomel, TJ, Lake, JP, and Comfort, P. Understanding the key phases of the countermovement jump force-time curve. *Strength Cond J* 40: 96–106, 2018.
- Meylan, C, McMaster, T, Cronin, J, Mohammad, NI, Rogers, C, and Dekker, M. Single-leg lateral, horizontal, and vertical jump assessment: Reliability, interrelationships, and ability to predict sprint and change-of-direction performance. *J Strength Cond Res* 23: 1140–1147, 2009.
- Sheppard, JM, Cormack, S, Taylor, K-L, McGuigan, MR, and Newton, RU. Assessing the force-velocity characteristics of the leg extensors in well-trained athletes: The incremental load power profile. *J Strength Cond Res* 19: 85–91, 2005.
- Slinde, F, Suber, C, Surer, L, Edwén, CE, and Svantesson, U. Test-retest reliability of three different countermovement jumping tests. *J Strength Cond Res* 22: 640–644, 2008.
- Suchomel, T and Stone, M. The relationships between hip and knee extensor cross-sectional area, strength, power, and potentiation characteristics. *Sports* 5: 66, 2017.
- Tavakoli, M and Dennick, R. Making sense of Cronbach's alpha. *Int J Med Educ* 2: 53–55, 2011.
- Vaverka, F, Jandačka, D, Zahradník, D, Uchytíl, J, Farana, R, Šupej, M, and et al. Effect of an arm swing on countermovement vertical jump performance in elite volleyball players: FINAL. *J Hum Kinet* 53: 41–50, 2016.

40. Walsh, MS, Böhm, H, Butterfield, MM, and Santhosam, J. Gender BIAS in the effects of arms and countermovement on jumping performance. *J Strength Cond Res* 21: 362–366, 2007.
41. Watkins, CM, Barillas, SR, Wong, MA, Archer, DC, Dobbs, IJ, Lockie, RG, et al. Determination of vertical jump as a measure of neuromuscular readiness and fatigue. *J Strength Cond Res* 31: 3305–3310, 2017.
42. Young, WB, Pryor, JF, and Wilson, GJ. Effects of instructions on characteristics of countermovement and drop jump performance. *J Strength Cond Res* 9: 232–236, 1995.
43. Ziv, G and Lidor, R. Vertical jump in female and male basketball players-A review of observational and experimental studies. *J Sci Med Sport* 13: 332–339, 2010.